Week 1: Random Variables & Discrete Distributions

Foundational Definitions:

Random Variables (RV): An RV is a function that assigns a numerical value to each outcome in a sample space.

Types: Discrete RV: Countable outcomes (# of heads coin flip)

Continuous RV: Uncountable outcomes (e.g. time, weight)

Sample space (Ω): Set of all possible outcomes of an experiment.

example: Die roll; $\Omega = 1,2,3,4,5,6$

Discrete probability distributions

Bernoulli Dist: A trial w/ only 2 outcomes, success(1) or failure(0)

PMF: $P(X=x) = p^x(1-p)^{1-x}$, $x \in 0,1$

Support: $x=0,1$ Mean: $\mathbb{E}[X]=p$ Variance: $Var(x)=p(1-p)$

Example: Tossing a fair coin: $p=0.5$, Success = heads.

Binomial Dist: Sum of n independent Bernoulli trials w/ same success probability p.

$X \sim Bin(n,p)$ · Support: $k=0,1,\dots n$

PMF $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$ Example: Tossing a coin 10 times, prob. of getting 4 heads.

Mean: np, Variance: $np(1-p)$

Poisson Dist: Counts # of events in fixed interval (time, space), given they occur independently

· $X \sim Poisson(\lambda)$

· PMF: $P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$ Support: $k=0,1,2\dots$

· Mean = Variance = λ example: # of emails received per hour.

Expectation & Variance (Discrete Case)

Expected value (Mean):

Def: $\mathbb{E}[X] = \sum_x x \cdot P(X=x)$

· For functions: $\mathbb{E}[g(X)] = \sum_x g(x) \cdot P(X=x)$

Variance:

$Var(x) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Moment Generating Functions (MGFs)

· MGF of discrete random variable X: $M_x(t) = E[e^{tX}] = \sum_x e^{tx} \cdot P(X=x)$

Properties:

· $M'_x(0) = \mathbb{E}[X]$ · $M''_x(0) = \mathbb{E}[X^2]$

Independence & Additivity

Independent RVs: X & Y are independent if: $P(X=x, Y=y) = P(X=x) \cdot P(Y=y)$

Additivity of Expectation: · Always holds: $\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

Additivity of Variance: Only if X & Y are independent: $Var(X+Y) = Var(X) + Var(Y)$

Properties & Applications

Distribution	Use case	Key Features
Bernoulli(p)	One trial, 2 outcomes	Base building block of Binomial
Binomial(n,p)	Repeated independent trials	Has fixed # of trials
Poisson(λ)	Rare event counts	Approximates Binomial for large n, small p.
Normal Dist.		

Week 2: Poisson, Exponential & Gamma Distributions

Poisson Dist. - In Depth Additive: if $X \sim Poisson(\lambda_1)$ & $Y \sim Poisson(\lambda_2)$ independently, then $X+Y \sim Poisson(\lambda_1 + \lambda_2)$

CONT

Exponential Distribution - In Depth

Def: Models time between successive events in a poisson process

Memoryless: $P(X > s+t \mid X > s) = P(X > t)$.

Probability Density Function (PDF): $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$

Key properties

· Mean: $\mathbb{E}[X] = \frac{1}{\lambda}$

· Variance: $Var(X) = \frac{1}{\lambda^2}$

MGF: $M_x(t) = \frac{\lambda}{\lambda - t}$ for $t < \lambda$

Gamma Distributions - In Depth

Definition: A generalization of the exponential distribution

· Sum of k independent exponential RVs w/ rate $\lambda = 1/\theta$

· $X \sim Gamma(k, \theta)$; shape $= k$, scale $= \theta$

PDF: $f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$, $x \geq 0$

Properties: Mean: $\mathbb{E}[X] = k\theta$, Variance: $Var(X) = k\theta^2$, $\Gamma(n) = (n-1)!$, when n is an int.

Relationships between Distributions:

· if $X \sim Exp(\lambda)$, then $X \sim Gamma(1, 1/\lambda)$

· if $X_1, \dots, X_k \overset{iid}{\sim} Exp(\lambda)$, then $\sum_{i=1}^{k} X_i \sim Gamma(k, 1/\lambda)$

Practical Applications:

Scenario	Best Model	Reason
Time between Arrivals	Exponential	Memoryless time between events
Time until k^{th} event	Gamma	Sum of k waiting times
Count of events in interval	Poisson	Rate λ over fixed time

Week 3: Conditional Expectation, Variance, & Joint Distributions

Conditional Probability:

Def: the cond. prob of A given B: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Conditional PMF/PDF:

· For discrete RVs: $P(X=x|Y=y) = \frac{P(X=x, Y=y)}{P(Y=y)}$

· For continuous RVs: $f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)}$

Conditional Expectation

Definition: Expectation of X given $Y=y$

· Discrete: $E[X|Y=y] = \sum_x x \cdot P(X=x|Y=y)$

· Continuous: $E[X|Y=y] = \int x f_{X|Y}(x|y)\,dx$

Properties

· $E[E[X|Y]] = E[X]$ (Law of iterated expectations)

· If X & Y are independent, $E[X|Y] = E[X]$

Joint Distributions

Joint PMF/PDF

Discrete: $P(X=x, Y=y)$

Continuous: $f_{X,Y}(x,y)$

Marginal Distributions

· Discrete: $P(X=x) = \sum_y P(X=x, Y=y)$

· Continuous: $f_X(x) = \int f_{XY}(x,y)\,dy$

Independence:

X & Y are independent iff:

· $P(X=x, Y=y) = P(X=x)P(Y=y)$

· $f_{XY}(x,y) = f_X(x) f_Y(y)$

Conditional Variance

Definition: $Var(X|Y=y) = E[(X - E[X|Y=y])^2 \mid Y=y]$

Law of total variance

$Var(X) = E[Var(X|Y)] + Var(E[X|Y])$

Practical Applications

Concepts	Use Case
$E[X|Y] = E[X] + E[Y]$	Prediction when info about Y is known
Law of Iterated Expectations	Decomposing hierarchical models
Conditional Variance	Uncertainty given known values
Law of Total Variance	Explaining spread in nested models

Examples

1) Conditional expectation from table

Given joint PMF $P(X=x, Y=y)$ in a table, compute

· $P(Y=y)$ marginal · $P(X=x|Y=y)$ conditional

· $E[X|Y=y]$

2) Iterated Expectation

Let $X|Y=y \sim Poisson(y)$ & $Y \sim Exp(1)$

· Compute $E[X] = E[E[X|Y]]$

Conditional Distributions & Sums

Sum of two RVs: expectation: $E[X+Y] = E[X] + E[Y]$

variance (if independent): $Var(X+Y) = Var(X) + Var(Y)$

Convolution for PMFs

· For $Z = X+Y$, $P_Z(z) = \sum_x P_X(x) P_Y(z-x)$

Week 4: Linear Model and Least Squares Estimation

Linear Models Overviews

Basic Form: linear model expresses a response variable Y as a linear function of predictors X: $Y = X\beta + \epsilon$

- Y: $n \times 1$ vector of responses
- X: $n \times p$ design matrix (each row is an observation)
- β: $p \times 1$ vector of coefficients
- ϵ: $n \times 1$ vector of random errors

Assumptions:

1) **Linearity:** Relationship between Y & X is linear in parameters
2) **Independence:** Observations are independent
3) **Homoscedasticity:** Constant variance of residuals $(Var(\epsilon_i) = \sigma^2)$
4) **Normality:** $\epsilon_i \sim N(0, \sigma^2)$ (needed for inference)

Least Squares Estimation

Goal: estimate β by minimizing residual sum of squares (RSS):
$$RSS = \|Y - X\beta\|^2 = (Y - X\beta)^T (Y - X\beta)$$

Solution
- Take derivative and solve $\hat{\beta} = (X^T X)^{-1} X^T Y$

With Intercept
- Add a column of 1s to X for intercept term. $X_{aug} = [1_n | X]$

Geometry of Least Squares

Projection: $\hat{Y} = X\hat{\beta}$ is the projection of Y onto the column space of X.
- **Residuals:** $r = Y - \hat{Y}$ orthogonal to X: $X^T r = 0$

Hat Matrix
- $H = X(X^T X)^{-1} X^T$ so that $\hat{Y} = HY$
- **Properties:** $H = H^T = H^2$ (symmetric and idempotent)

Residuals and Models fits

- **Residuals:** $r_i = Y_i - \hat{Y}_i$ • **Properties:** $\sum r_i = 0$ (intercepts included)

R-squared (R^2)
- Proportion of variance explained by the model:
$$R^2 = 1 - \frac{RSS}{TSS} = \frac{\text{Explained Variance}}{\text{Total Variance}}$$

Inference for β

Sampling Distribution: if errors $\epsilon \sim N(0, \sigma^2 I)$, then $\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})$

Std. Error & Confidence Interval
$$SE(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X^T X)^{-1}]_{jj}}$$
- $\hat{\sigma}^2$: estimated variance of residuals
- $[(X^T X)^{-1}]_{jj}$: the j-th diagonal element of the inverse mtx
- Confidence interval (CI): $\hat{\beta}_j \pm z_{\alpha/2} \cdot SE(\hat{\beta}_j)$
  - Use $z_{\alpha/2}$ (normal dist.) for large samples
  - Use $t_{\alpha/2, n-p}$ (t-distribution) is estimated & small

Hypothesis Test (for coefficient β_j)
- **Null Hypothesis:** $H_0: \beta_j = 0$
- **Test Statistic** $t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$
  - follows a t-distribution with $n-p$ degrees freedom
  - Compares critical t value or compute p-value.

Multiple Linear Regression

- **Extension:** multiple predictors: $X_1, \ldots X_p$
- **Model:** $Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p + \epsilon$

Interpretation
- β_j measures effect X_j holding others

Model Diagnostics

Residual plots: plot residuals vs. fitted values to check
- Non-linearity • Unequal variance (heteroscedasticity) • Outliers

Q-Q Plot: checks normality assumption of residuals

Summary table:

Concepts	Formula
Least Squares Estimator	$\hat{\beta} = (X^T X)^{-1} X^T Y$
Fitted values	$\hat{Y} = X\hat{\beta}$
Residuals	$r = Y - \hat{Y}$
Hat Matrix	$H = X(X^T X)^{-1} X^T$
R-squared	$1 - \frac{RSS}{TSS}$
$Var(\hat{\beta})$	$\sigma^2 (X^T X)^{-1}$

Week 5: Poisson Generalized Linear Models.

Generalized linear models (GLMs) overview

Structure of a GLM: 3 main components
- i) **Random component:** Response variable Y from a dist. in the exponential family (e.g) Poisson, Binomial, Normal)
- ii) **Systematic Component** Linear Predictor $\eta = X\beta$
- iii) **Link function:** function g s.t. $g(\mu) = \eta$, where $\mu = E[Y]$

GLM Framework summary:
- Y_i - Response Variable
- μ_i - Mean of Y_i, $\mu_i = E[Y_i]$
- $g(\cdot)$ - Link function
- η_i - Linear Predictor

Log-Likelihood Function

Poisson log-Likelihood:

Given $Y_i \sim Poisson(\mu_i = \exp(X_i^T \beta))$:
$$\ell(\beta) = \sum_{i=1}^{n} [y_i x_i^T \beta - \exp(x_i^T \beta) - \log(y_i!)]$$

Score & information
- **Score function (gradient):** $\nabla_\beta \ell(\beta)$
- **Fisher information:**

$I(\beta) = X^T W X$, where $W_i = \mu_i$

Mean-Variance Relationship

Poisson: variance = Mean: $Var(Y_i) = \mu_i$

Overdispersion:
When observed variance > predicted variance

Fix: Use Neg Binomial model?

Model Diagnostics

Residuals: Deviance Residual:
$$r_i = \text{sign}(y_i - \hat{\mu}_i) \cdot \sqrt{2\left(y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) - (y_i - \hat{\mu}_i)\right)}$$

Pearson Residual: $r_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$

Applications

Scenario	Use Poisson GLM when...
Counting disease incidence	Counts per population or time
Modeling web traffic	Number of visits per time unit
Ecology	Number of animals in a region

Poisson GLM

Distribution: $Y \sim Poisson(\mu_i)$,

PMF: $P(Y_i = y_i) = \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!}$, $y \in 0, 1, 2, 3 \ldots$

Link function: **Canonical link:** $g(\mu) = \log(\mu)$

Model becomes: $\log(\mu_i) = x_i^T \beta \Rightarrow \mu_i = \exp(X_i^T \beta)$

Fitting the Model

Iteratively Reweighted Least Squares (IRLS)
- Used to solve β that maximizes $\ell(\beta)$
- Updates the of the form:
$$\beta^{(t+1)} = (X^T W X)^{-1} X^T W z$$
  - W: weights from current μ
  - z: adjusted dependent variable (working response)

Software: use statsmodels.api.GLM or sklearn.linear_model.PoissonRegressor in Python

Summary table:

Term	Meaning
μ_i	Expected count for observation i
$\log(\mu_i)$	Linear predictor
$\ell(\beta)$	Log-likelihood
IRLS	Algorithm to fit GLM
Canonical Link	log for Poisson

Week 6: Gamma GLMs, Overdispersion, & Model Fit

Gamma dist. overview

- Def: Continuous dist. for strictly positive data
- Commonly used when the response variable is right skewed & strictly pos (e.g. waiting time, insurance cost, rainfall)
- $Y \sim \text{Gamma}(\alpha, \theta)$ • α: shape param • θ scale param.

PDF: $f(y; \alpha, \theta) = \frac{1}{\Gamma(\alpha)\theta^{\alpha}} y^{\alpha-1} e^{-y/\theta}, y > 0$

Properties: • Mean: $E[Y] = \alpha\theta$ • $\text{Gamma}(1, \theta) = \text{Exponential}(\lambda = 1/\theta)$

• **Variance:** $\text{Var}(Y) = \alpha\theta^2$

Gamma Generalized Linear Model (GLM)

- **When to use:** Response var is positive & continuous
- When variable increases w/ square of the mean (ie, $\text{Var}(Y_i) \propto \mu_i^2$)

Link Function

- **Common:** Log link $\Rightarrow \log(\mu_i) = X_i^T \beta \Rightarrow \mu_i = e^{X_i^T \beta}$
- **Alternatives:** Inverse link $\Rightarrow g(\mu) = 1/\mu$

Gamma GLM structure:

- **Distribution:** $Y_i \sim \text{Gamma}(\alpha, \theta_i)$
- **Systematic** part: $\eta_i = X_i^T \beta$
- **Mean:** $\mu_i = E[Y_i] = g^{-1}(\eta_i)$

Log Likelihood & Estimation

Log-Likelihood: For log link function & $\mu_i = e^{X_i^T \beta}$, the **log-likelihood becomes:** $\ell(\beta) = \sum_{i=1}^{n} \left[\alpha\log(\alpha) - \log(\Gamma(\alpha)) - \alpha\log(\mu_i) - \frac{\alpha y}{\mu_i} \right]$

Estimation Method:

- Use iteratively reweighted least squares (IRLS) to maximize log-likelihood
- most software libraries do this automatically

Residuals for Gamma GLMs

Deviance Residual: $r_i = \text{sign}(y_i - \mu_i) \cdot \sqrt{2\left[\frac{y_i - \mu_i}{\mu_i} - \log\left(\frac{y_i}{\mu_i}\right)\right]}$

Pearson Residual: $r_i = \frac{y_i - \mu_i}{\mu_i}$

Overdispersion in GLMs

Definition - overdispersion occurs when: Observed Variance > Theoretical var.

Fix: Use quasi-likelihood methods

- switch from poisson to neg-binomial or Gamma

Model Selection & (AIC)

Akaike Information Criterion (AIC)

$AIC = 2k - 2\log(\hat{L})$ • Lower AIC indicates better model (penalizes complexity)

Compare Models

- Compare Gamma vs Poisson vs. Gaussian for continuous skewed data • Use cross validation or AIC/BIC

Application Scenarios

Context	Why Gamma GLM
Insurance claims	Positive, right skewed data
Cost Modeling	Heteroscedastic continuous response
Engineering Failure Times	Time until failure; skewed data

Summary Table

- μ_i - Expected mean of Y_i
- **Log link** - $g(\mu_i) = \log(\mu_i)$
- $\text{Var}(Y_i)$ - $\alpha\mu_i^2$
- **Use gamma when** - Response is (+) & heteroscedastic

Week 7: Model Eval - AIC, BIC, Residuals, & Diagnostics

Model Fit & Deviance

Deviance: measures the goodness of fit for GLMs by comparing the fitted model to a saturated model (perfect fit).

Formula: $D = 2[\ell(\text{saturated model}) - \ell(\text{fitted model})]$

Null deviance vs Residual deviance

- **Null deviance:** Deviance from a model w/ only the intercept (no predictors)
- **Residual deviance:** Deviance from the model w/ all predictors
- Lower residual deviance means better fit.

Information Criteria:

AIC - Akaike Information Criterion: $AIC = 2k - 2\log(\hat{L})$

- **k**: # of parameters in the model
- $\hat{L}$: maximized value of the likelihood function.

BIC: Bayesian Information Criterion: $BIC = k\log(n) - 2\log(\hat{L})$

- **n**: # of observations
- BIC penalizes complexity of models more than AIC.

Use cases: AIC better for prediction accuracy

BIC better for model selection under a true model framework.

Residuals in GLMs

Type	Formula	Interpretation
Raw	$y_i - \hat{\mu}_i$	Basic Residual
Pearson	$\frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$	Standardized Residual
Deviance	Based on Likelihood	Measures model fit
Anscombe	Uses transformation	More symmetry in skewed models

Plotting Residuals

- Plot residuals vs fitted values to check:
  - Heteroscedasticity
  - Non-linearity
  - Outliers

Model Diagnostics

Key plots: Residuals vs Fitted: Should show no pattern (checks linearity, equal variance)

QQ Plot: Should be approx. linear if residuals are normal

Scale location plot: Spread of residuals should be constant

Leverage Plot: Identifies influential observations

Leverage & Cook's distance:

- Leverage measures influence of data points x_i on fitted $\hat{y}_i$.
- Cooks Distance combines leverage & residuals: $D_i = \frac{(r_i^2 h_{ii})}{p \cdot MSE(1-h_{ii})^2}$

Overdispersion Detection

Pearson Chi-Square statistic: $\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^{n} \left(\frac{y_i - \mu_i}{\sqrt{V(\mu_i)}}\right)^2$

- Estimate overdispersion factor
- $\hat{\phi} > 1$ suggests overdispersion

Consequences of Ignoring Overdispersion

- Underestimated std. errors $\rightarrow$ misleading inference
- Poor model fit & invalid prediction intervals

Comparing nested models

Likelihood Ratio Test:

- Compare 2 nested models: $D = 2(\ell_{full} - \ell_{reduced})$
- $D \sim \chi^2_{df}$ under the null hypothesis

• **Degrees of freedom** = diff in # of param.

Cross Validation

K-Fold CV

- Partition data into k subsets, train on k-1 & test on the 1 left out.
- Rotate & average test errors

Use in GLMs: Evaluate out of sample performance

- Useful when comparing models w/ similar AIC/BIC.

Summary Table

Metric	Formula	Usage
AIC	$2k - 2\log\hat{L}$	Prediction focused model comparison
BIC	$k\log(n) - 2\log\hat{L}$	Penalizes complexity more, better for true model
Deviance	$2(\ell_{sat} - \ell_{fit})$	Goodness-of-fit
Pearson χ^2	$\sum\left(\frac{y-\mu}{\sqrt{V(\mu)}}\right)^2$	Detect overdispersion
Cook's D	Measures Influence	Outlier + leverage detection

Week 8: Logistic Regression & Bernoulli Models

Binary outcome modeling

- When to use logistic regression: Response var $Y_i \in 0,1$ (e.g. success/failure, yes/no, win/loss)
- Models the prob. of success as a function of predictors

Bernoulli Distribution

PMF: $P(Y=y) = p^y(1-p)^{1-y}, y \in 0,1$

Properties:
$E[Y] = p$, $Var(Y_i) = p(1-p)$

GLM form
- $Y_i \sim \text{Bernoulli}(p_i)$
- Canonical link: logit

Logistic Regression Model

Model: $\log\left(\frac{p_i}{1-p_i}\right) = X_i^T\beta$ (log-odds)

- Inverse link (sigmoid function): $p_i = \frac{e^{X_i^T\beta}}{1+e^{X_i^T\beta}} = \frac{1}{1+e^{-X_i^T\beta}}$

Interpretation of Coeff.
- A one unit increase in X_j leads to change in the log-odds of outcome by β_j
- Odds ratio: $\exp(\beta_j)$ = multiplicative change in odds per unit increase in X_j

Log-Likelihood

Bernoulli Log-Likelihood

$\ell(\beta) = \sum_{i=1}^{n}\left[y_i\log(p_i) + (1-y_i)\log(1-p_i)\right]$

- Concave function: guarantees a global maximum
- Solved using gradient-based optimization (e.g., Newton-Raphson or IRLS)

Estimation via IRLS

Iteratively Reweighted Least Squares (IRLS)

- At each step:
  - Compute $p_i = \frac{1}{1+e^{-x_i^T\beta}}$
  - Construct weights $W_i = p_i(1-p_i)$
  - Update: $\beta^{(t+1)} = (X^TWX)^{-1}X^TWz$ where z is the adjusted dependent variable

Model Diagnostics

Deviance Residuals

- Used to assess fit for logistic models
- Lack-of-fit if residuals are large or asymmetrically distributed

Hosmer-Lemeshow Test

- Compare predicted probabilities to observed outcomes grouped by decile

Multicollinearity

- Check Variance Inflation Factors (VIFs)

Classification Thresholds

- Default threshold: 0.5
- ROC curve: plot of TPR vs. FPR at various thresholds
- AUC (Area Under Curve): Measures overall classifier quality

Confusion Matrix Terms

Term	Meaning
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Performance Metrics

- Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$
- Precision: $\frac{TP}{TP+FP}$
- Recall (Sensitivity): $\frac{TP}{TP+FN}$
- F1 Score: Harmonic mean of precision and recall

Applications of Logistic Regression

Application	Use Case
Medical Diagnosis	Disease vs. No Disease
Email Filtering	Spam vs. Not Spam
Credit Risk	Default vs. No Default
Marketing	Response vs. No Response

Summary Table

Concept	Formula / Meaning
Link Function	$\log\left(\frac{p}{1-p}\right)$
Inverse Link	$p = \frac{e^\eta}{1+e^\eta}$
Log-likelihood	$\ell(\beta) = \sum[y_i\log(p_i) + (1-y_i)\log(1-p_i)]$
IRLS Update	$\beta^{(t+1)} = (X^TWX)^{-1}X^TWz$
Coeff. Interpretation	$\exp(\beta_j)$ = odds ratio

Week 9: Inference for GLMS - Hypothesis Testing, Conidence Intervals, & Significance

Goals of Statistical Inference

- Assess which predictors are significant in explaining variation in response
- Compute **confidence intervals** for model parameters β
- Perform **hypothesis testing** on regression coefficients

Standard Errors & Confidence Intervals

Variance of Coefficient Estimates

$$\mathrm{Var}(\hat{\beta}) \approx (X^TWX)^{-1}$$

- W: weight matrix from IRLS (depends on fitted values)

Standard Error (SE)

$$\mathrm{SE}(\hat{\beta}_j) = \sqrt{\left[\mathrm{Var}(\hat{\beta})\right]_{jj}}$$

- This is the square root of the **j-th diagonal** element of the variance matrix.

Confidence Interval (CI)

$$\hat{\beta}_j \pm z_{\alpha/2} \cdot \mathrm{SE}(\hat{\beta}_j)$$

- For **95% CI**, use $z = 1.96$
- This gives a range of plausible values for each coefficient.

Asymptotic Properties of MLEs

Maximum Likelihood Estimator (MLE)

- Under regularity conditions, MLEs are:
  - Consistent: $\hat{\beta} \xrightarrow{p} \beta$
  - Asymptotically normal: $\hat{\beta} \sim N\left(\beta, I(\beta)^{-1}\right)$

Fisher Information

- $I(\beta)$ is the expected curvature of the log-likelihood: $I(\beta) = -E[\nabla^2\ell(\beta)]$
- Estimated by observed information matrix from second derivative

Hypothesis Testing

Wald Test (Z-Test)

- Test $H_0: \beta_j = 0$
- Test statistic: $z_j = \frac{\hat{\beta}_j}{\mathrm{SE}(\hat{\beta}_j)} \sim N(0,1)$
- p-value $= 2(1 - \Phi(|z_j|))$

Likelihood Ratio Test (LRT)

- Compare full model vs. reduced model (dropping one or more coefficients)
- Test statistic: $D = 2\left[\ell_{\text{full}} - \ell_{\text{reduced}}\right] \sim \chi^2_{df}$

Score Test (Lagrange Multiplier Test)

- Uses derivative of likelihood at null value

Multiple Coefficient Tests

Joint Significance Testing

- Null: $H_0: \beta_1 = \beta_2 = \cdots = \beta_k = 0$
- Use LRT or F-test analog for GLMs

Type I vs Type II Error

- Type I: Rejecting H_0 when it is true (false positive)
- Type II: Failing to reject H_0 when it is false (false negative)

p-values and Interpretation

- Small p-value (< 0.05): Evidence against H_0
- Large p-value (> 0.05): Fail to reject H_0
- Always interpret in context of model and domain

Summary Table

Test	Statistic	Distribution	Use
Wald	$z = \hat{\beta}\ /\ \mathrm{SE}$	$N(0,1)$	Single coefficient
LRT	$D = 2(\ell_{\text{full}} - \ell_{\text{reduced}})$	χ^2	Nested model comparison
Score	Based on gradient	χ^2	Efficient null hypothesis check

Week 10: Final Review - Applications, Summary & Model Comparison

Review of Core Models

Linear Model (LM)

- Response: continuous
- Form: $Y = X\beta + \epsilon$
- Assumes: Gaussian errors, constant variance
- Estimation: Ordinary Least Squares (OLS)

Logistic Regression

- Response: binary (0 or 1)
- Link: logit — $\log\left(\frac{p}{1-p}\right)$
- Output: predicted probabilities
- Use case: classification

Poisson Regression

- Response: counts (integers ≥ 0)
- Link: log — $\log(\mu)$
- Assumes: mean = variance (Poisson)
- Use case: event counts (e.g. calls, claims)

Gamma Regression

- Response: continuous, positive, right-skewed
- Link: log or inverse
- Variance increases with square of the mean
- Use case: cost, rainfall, failure time

Model Selection & Use Cases

Model	Response Type	Link	Use Case
Linear	Continuous	Identity	Height, temperature
Logistic	Binary	Logit	Classification: spam, disease
Poisson	Count	Log	Number of calls, claims
Gamma	Positive continuous	Log / Inverse	Time, money, risk

Model Comparison Techniques

AIC and BIC

- AIC: $2k - 2\log(\hat{L})$ (best for prediction)
- BIC: $k\log(n) - 2\log(\hat{L})$ (best for model selection)

Likelihood Ratio Test (LRT)

- Compare nested models
- $D = 2(\ell_{\text{full}} - \ell_{\text{reduced}}) \sim \chi^2$

Cross-Validation (CV)

- K-fold CV: rotate test/train data
- Use prediction error as selection criterion

Diagnostic Checks

Residuals

- Raw: $y_i - \mu_i$
- Pearson: scaled by variance
- Deviance: model-specific, compares to saturated model

Plots

- Residual vs Fitted: checks linearity & variance
- Q-Q Plot: checks normality assumption
- ROC Curve: classification performance

Overdispersion

- Use Pearson Chi-squared test statistic
- Use Negative Binomial or Quasi-GLM for correction

General Strategy for Model Building

1. **Understand your outcome**: binary? count? continuous?
2. **Explore data**: histogram, boxplot, scatterplot
3. **Choose appropriate model family and link**
4. **Fit and check assumptions**
5. **Use AIC/BIC/CV to compare models**
6. **Interpret coefficients in context**

Interpretation of Coefficients (Quick Review)

- **Linear**: β_j is the change in Y per unit change in X_j
- **Logistic**: $\exp(\beta_j)$ is the odds ratio for a unit increase in X_j
- **Poisson**: $\exp(\beta_j)$ is the multiplicative effect on expected count
- **Gamma**: $\exp(\beta_j)$ is the multiplicative effect on expected mean